

MUHAMMAD FADI S

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Education

Vellore Institute of Technology, Chennai

B.Tech in Electronics & Communication Engineering

2024 – 2028

CGPA: 9.14/10

Delhi Public School, Electronic City

Higher Secondary (PCMC with Computer Science), CBSE

2022 – 2024

92.4%

Relevant Skills

Analog & RF Electronics: Op-Amps, Oscillators, Amplifiers, Modulation/Demodulation Circuits

EDA & Digital Design: Cadence Virtuoso, LTSpice, Ansys HFSS (learning), Verilog HDL, FPGA (Quartus Prime, Vivado), Keil µVision

Embedded Systems: Arduino, ESP32, Raspberry Pi, MATLAB, Sensor Integration

Robotics & Systems: ROS2, PX4, ArduPilot, QGroundControl

Programming: Python, Java, Assembly (8051), Perl (learning)

Projects

Continuous Time Linear Equalizer for 12.5Gb/s Links | Cadence Virtuoso

Aug 2026

- Designed and simulated a two-stage CTLE in 45nm technology to mitigate Inter-Symbol Interference across orthogonal and traditional backplanes, achieving ~10 dB peaking gain.

FPGA-Based PCB Defect Detection | Zynq Z2, YOLOv8

In Progress

- Designing a YOLOv8s-based real-time defect detection architecture in Verilog HDL, targeting a **Xilinx Zynq Z2 FPGA/SoC**; currently verified via Proof of Concept RTL of first layer.
- Planning hardware-software co-design using Vivado and PYNQ for on-board inference in electronics manufacturing quality inspection.”

SIW Bandpass Filter for X-Band Applications | Ansys HFSS

In Progress

- Designing a Half Mode Substrate Integrated Waveguide (SIW) bandpass filter with Complementary Split Ring Resonators (**CSSR**) for X-band, modeling via-wall geometry and coupling structures for a target passband in Ansys HFSS.

4×4 Array Multiplier using Hybrid TGL Logic | Cadence Virtuoso

Mar 2026

- Designed schematic, layout, and post-layout simulations using hybrid Transmission Gate Logic; reduced transistor count by 25% (392→292) vs. conventional CMOS.
- Achieved lower propagation delay and improved power efficiency through optimized layout-level design.

Analog Voltage Controlled Oscillator (VCO) with 4-Step Sequencer

Nov 2025

- Designed a VCO using a **CD40106** Schmitt-trigger oscillator with a thermally coupled **PNP-NPN exponential control stage** for stable frequency control.
- Built a **4-step voltage sequencer** using a 555 Timer IC, generating **7.12 kHz** stepped frequency output.

Experience

Industrial Trainee

Jun 2026 – Jul 2026

Times Microwave Systems (An Amphenol Company), Electronic City, Bangalore

- Learned the full lifecycle of high-reliability RF cable assemblies, including manufacturing, cutting, stripping, connector assembly, soldering, inspection, and qualification for aerospace, defense, and telecom interconnects.
- Gained exposure to RF testing via **VNA** measurements, S-parameters, VSWR, insertion loss, **TDR**, and **PIM**; observed thermal cycling, Hi-Pot testing, and QA/failure-analysis procedures.

Avionics Member

Apr 2025 – Present

Team Aviators International, College Drone Team, VIT Chennai

- Led component selection for power, payload, and endurance at **SAEISS DDC 2026**, placing **3rd overall** out of 45 teams.
- Placed **4th out of 17** teams at Dronetrix, **IIST Trivandrum**; designed, calibrated, and assembled two high-velocity FPV racing drones.
- Configured sensors via **QGroundControl** & **Mission Planner** on **PX4** & **ArduPilot**; analyzed thrust-to-weight and power requirements.

Achievements

- Branch Topper**, Electronics & Communication Engineering, VIT Chennai (2024–2025, First Year)
- SAEISS DDC 2026**: 3rd out of 45 teams, UAV design competition.
- Dronetrix, IIST UAV Competition (2025)**: 4th out of 17 teams in FPV racing drone event.
- Smart India Hackathon (2025)**: Selected from 800+ college teams for 24-hour internal hackathon.